

Chapter 2

Discrete Probability Distributions

In Chapter 1 we described data we had already collected. Now we turn to a different question: before we collect any data, can we describe what we *expect* to see?

A **probability distribution** is a complete description of all possible outcomes of a random process and how likely each one is. We will represent these distributions in three equivalent ways: as a *table*, as a *formula*, and as a *histogram*. All three say the same thing; each one reveals something different.

The histogram is the most important. Every probability distribution in this course — discrete or continuous — lives as a picture. The height of a bar, or the height of a curve, represents how likely that outcome is. The total area of the picture always equals exactly 1. Keep that image in mind as you work through the rest of the course.

2.1 Random Variables

Definition: Random Variable. A **random variable** is a variable whose value is determined by the outcome of a random process. We typically denote random variables with capital letters (X, Y). A specific observed value is written in lowercase (x).

Definition: Discrete Random Variable. A random variable is **discrete** if it can take only a countable number of values — usually whole numbers. Examples: number of heads in 10 coin flips, number of customers per hour, number of defective items in a shipment.

Definition: Probability Distribution (discrete). The **probability distribution** of a discrete random variable X assigns a probability $P(X = x)$ to each possible value x . Two rules must hold:

1. $P(X = x) \geq 0$ for all x (probabilities are non-negative).

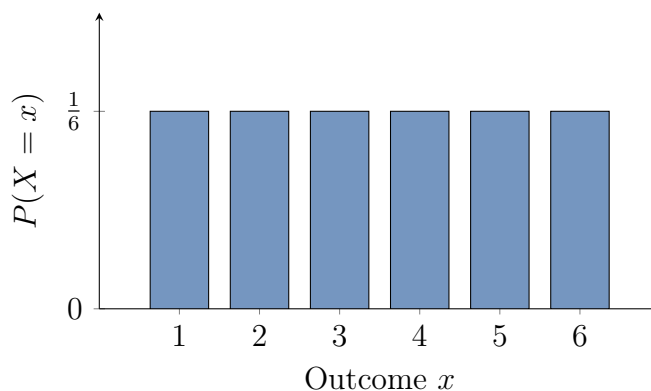
$$2. \sum_{\text{all } x} P(X = x) = 1 \text{ (probabilities sum to 1).}$$

Visual Insight. Rule 2 is the discrete version of a fact that runs through the entire course: *the total probability is always 1*. For a discrete distribution, this means the bar heights in the probability histogram sum to 1. Later, for continuous distributions, it means the total **area under the curve** equals 1.

Example 2.1.1. (*A Simple Probability Distribution*) Roll a fair six-sided die. Let X be the result. Each face has probability $\frac{1}{6}$.

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Probability Histogram: Fair Die



Every bar has the same height. Six bars, each of height $\frac{1}{6}$, total = 1.

Exercises 2.1

2.1.1. A bag contains 3 red, 2 blue, and 1 green marble. You draw one marble at random. Let X be the color (coded 1 = red, 2 = blue, 3 = green). Write the probability distribution as a table and verify that the probabilities sum to 1.

2.1.2. Is the following a valid probability distribution? Why or why not?

x	0	1	2	3
P	0.2	0.4	0.5	-0.1

2.1.3. A spinner has four sectors with probabilities 0.3, 0.3, 0.2, and 0.2. Sketch the probability histogram.

2.2 Expected Value and Variance

2.2.1 Expected Value

The **expected value** of a random variable is its long-run average — what we would get if we repeated the random process many times and took the mean of the results.

Key Formula. The **expected value** (or **population mean**) of a discrete random variable X is:

$$\mu = E(X) = \sum_{\text{all } x} x \cdot P(X = x).$$

Each value x is weighted by its probability. More likely values contribute more to the average.

Notice the notation: μ (Greek mu) for the expected value, because it is a *population* parameter — a theoretical quantity, not computed from sample data.

Example 2.2.1. (*Expected Value of a Die Roll*) For a fair die, $\mu = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{21}{6} = 3.5$.

We do not expect to actually roll a 3.5 — it is impossible. The expected value is a long-run average, not a single predicted outcome.

Example 2.2.2. (*Expected Value from a Table*) An insurance company offers a policy. If the policyholder files a claim, the company pays \$5,000; if no claim is filed (probability 0.97), it pays nothing. Let X be the company's payout.

Outcome	x	$P(X = x)$
No claim	0	0.97
Claim	5000	0.03

$$\mu = 0(0.97) + 5000(0.03) = \$150.$$

The company expects to pay out \$150 per policy on average, so it must charge more than \$150 per policy to be profitable.

2.2.2 Variance and Standard Deviation

Key Formula. The **variance** of a discrete random variable is:

$$\sigma^2 = \sum_{\text{all } x} (x - \mu)^2 \cdot P(X = x).$$

The **standard deviation** is $\sigma = \sqrt{\sigma^2}$.

Both use σ (Greek sigma) — population parameters.

Compare to Chapter 1: the sample variance was $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$. Here, instead of dividing by $n - 1$, we weight each squared deviation by its probability.

Example 2.2.3. (*Variance of a Fair Die*) $\mu = 3.5$. Computing $(x - 3.5)^2 \cdot \frac{1}{6}$ for each face:

x	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^2 \cdot P$
1	-2.5	6.25	1.0417
2	-1.5	2.25	0.3750
3	-0.5	0.25	0.0417
4	+0.5	0.25	0.0417
5	+1.5	2.25	0.3750
6	+2.5	6.25	1.0417
$\sigma^2 = 2.9167$			

$$\sigma = \sqrt{2.9167} \approx 1.708.$$

Exercises 2.2

2.2.1. A game costs \$2 to play. You roll one die: if you roll a 6, you win \$10; otherwise you win nothing. Let X be your *net* gain (winnings minus cost). Build the distribution table and compute μ . Is this game fair?

2.2.2. Verify that $\sum (x - \mu)^2 P(X = x)$ for the insurance example (Example 2.2.2) equals 727,500.

2.2.3. Two investments: Investment A has $\mu = \$500$, $\sigma = \$50$. Investment B has $\mu = \$500$, $\sigma = \$400$. Which would you prefer and why? Is there a single correct answer?

2.3 The Geometric Distribution

2.3.1 The Setting: Waiting for the First Success

Many situations ask: *How long until the first success?*

- How many times do you flip a coin before getting the first head?
- How many free throws does a player attempt before making the first one?
- How many parts does an inspector check before finding the first defect?

These are all **geometric** situations. They share a common structure:

1. Each trial is independent.
2. Each trial has exactly two outcomes: success or failure.
3. The probability of success p is the same on every trial.
4. We count the number of trials until (and including) the first success.

Definition: Geometric Distribution. If X is the number of trials until the first success, with success probability p on each trial, then X has a **geometric distribution** and:

$$P(X = k) = (1 - p)^{k-1} \cdot p, \quad k = 1, 2, 3, \dots$$

2.3.2 Reading the Formula

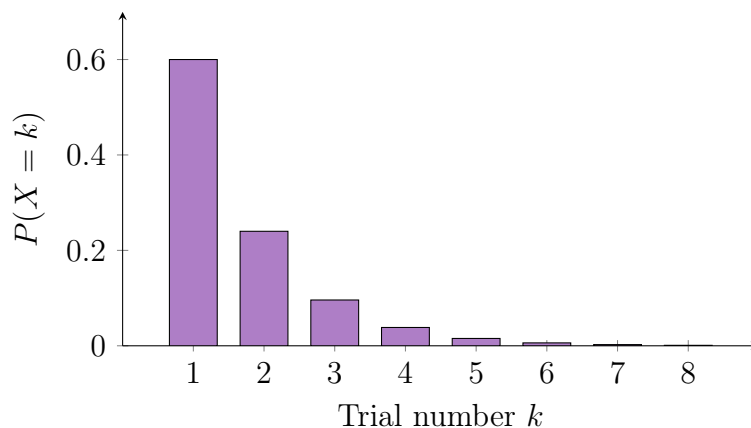
$(1-p)^{k-1}$ is the probability of failing $k-1$ times in a row. p is the probability of succeeding on the k -th try. Because the trials are independent, we multiply.

Example 2.3.1. (Free Throws) A basketball player makes 60% of her free throws ($p = 0.60$). What is the probability that she first makes a free throw on her 3rd attempt?

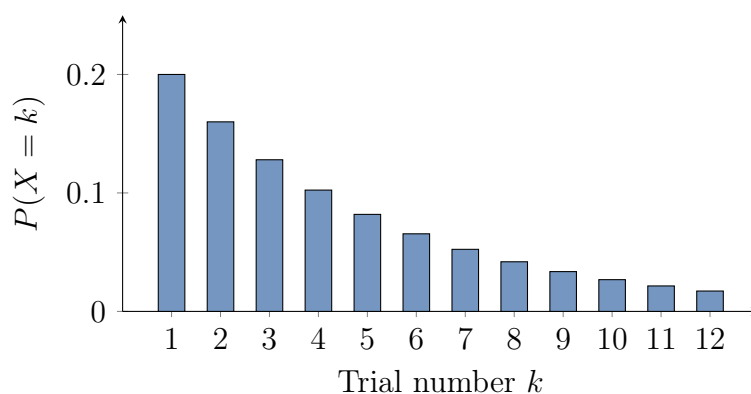
$$P(X = 3) = (1 - 0.60)^{3-1} \cdot 0.60 = (0.40)^2 \cdot 0.60 = 0.16 \cdot 0.60 = 0.096.$$

About a 9.6% chance.

Visual Insight. The geometric distribution always looks the same: a tall bar at $k = 1$ that decreases as k increases. The bars never reach zero — there is always some probability you need one more try — but they shrink fast. Here is the picture for $p = 0.60$ (the free throw example):

Geometric Distribution: $p = 0.60$ 

Compare $p = 0.60$ to $p = 0.20$: a lower success probability means many more trials are likely needed, so the distribution spreads out to the right.

Geometric Distribution: $p = 0.20$ 

When p is small, success is rare, and the histogram has a long tail to the right — just like the skewed-right distributions from Chapter 1.

2.3.3 Mean and Standard Deviation of Geometric

Key Formula. For a geometric distribution with success probability p :

$$\mu = \frac{1}{p}, \quad \sigma = \frac{\sqrt{1-p}}{p}.$$

The mean $\frac{1}{p}$ has a beautiful interpretation: if success probability is p , you expect to

need $\frac{1}{p}$ trials on average. If $p = 0.5$, you expect 2 trials. If $p = 0.1$, you expect 10 trials. If $p = 0.01$, you expect 100 trials.

Example 2.3.2. (*Expected Waiting Time*) The free throw player ($p = 0.60$):

$$\mu = \frac{1}{0.60} \approx 1.67 \text{ attempts.}$$

She expects to need fewer than 2 attempts to make her first free throw, which makes sense — she makes 60% of them.

For $p = 0.20$:

$$\mu = \frac{1}{0.20} = 5 \text{ trials.}$$

This matches the histogram above: the center of mass of those bars is around $k = 5$.

Exercises 2.3

2.3.1. A manufacturing process produces defective items with probability 0.08. Parts are inspected one at a time. Let X = number of parts inspected until the first defect. (a) Compute $P(X = 1)$, $P(X = 2)$, $P(X = 3)$. (b) Compute μ and σ . (c) Sketch the first 6 bars of the probability histogram.

2.3.2. A student guesses randomly on a multiple-choice exam where each question has 4 choices. What is the probability the student gets the first correct answer on question number 5?

2.3.3. Show algebraically that $\sum_{k=1}^{\infty} P(X = k) = 1$ for the geometric distribution. (*Hint:* use the formula for an infinite geometric series $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ for $|r| < 1$.)

2.4 The Binomial Distribution

2.4.1 The Setting: Counting Successes in Fixed Trials

The geometric distribution counted *trials until* the first success. The binomial distribution counts *how many successes* in a fixed number of trials.

Definition: Binomial Setting. A situation is a **binomial setting** if:

1. There are exactly n trials (fixed in advance).
2. Each trial results in success or failure.
3. The probability of success p is the same on every trial.
4. The trials are independent.

The random variable $X = \text{number of successes in } n \text{ trials}$ has a **binomial distribution**, written $X \sim \text{Bin}(n, p)$.

2.4.2 Counting Arrangements: The Combination Formula

Before we can write the binomial probability formula, we need to count how many ways k successes can occur among n trials.

Definition: Combination. The number of ways to choose k items from n (without regard to order) is:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Read “ n choose k .” Here $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$ (“ n factorial”), and $0! = 1$ by convention.

Example 2.4.1. (*Computing Combinations*) How many ways can exactly 2 heads appear in 4 coin flips?

$$\binom{4}{2} = \frac{4!}{2! \cdot 2!} = \frac{24}{2 \cdot 2} = 6.$$

The six arrangements are: HHTT, HTHT, HTTH, THHT, THTH, TTHH. Count them: yes, 6.

2.4.3 The Binomial Probability Formula

Key Formula. If $X \sim \text{Bin}(n, p)$, the probability of exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n.$$

- $\binom{n}{k}$: the number of arrangements of k successes among n trials.
- p^k : the probability of those k trials all being successes.
- $(1-p)^{n-k}$: the probability of the remaining $n-k$ trials all being failures.

We multiply because the trials are independent.

Example 2.4.2. (*Binomial Probability*) A student guesses randomly on a 10-question true/false quiz ($n = 10, p = 0.5$). What is the probability of getting exactly 7 correct?

$$P(X = 7) = \binom{10}{7} (0.5)^7 (0.5)^3 = 120 \cdot (0.5)^{10} = 120 \cdot \frac{1}{1024} \approx 0.1172.$$

About an 11.7% chance.

Example 2.4.3. (*Cumulative Probability*) Same quiz. What is the probability of passing (7 or more correct)?

$$P(X \geq 7) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10).$$

$$P(X = 7) = \binom{10}{7} (0.5)^{10} = 120/1024 \approx 0.1172$$

$$P(X = 8) = \binom{10}{8} (0.5)^{10} = 45/1024 \approx 0.0439$$

$$P(X = 9) = \binom{10}{9} (0.5)^{10} = 10/1024 \approx 0.0098$$

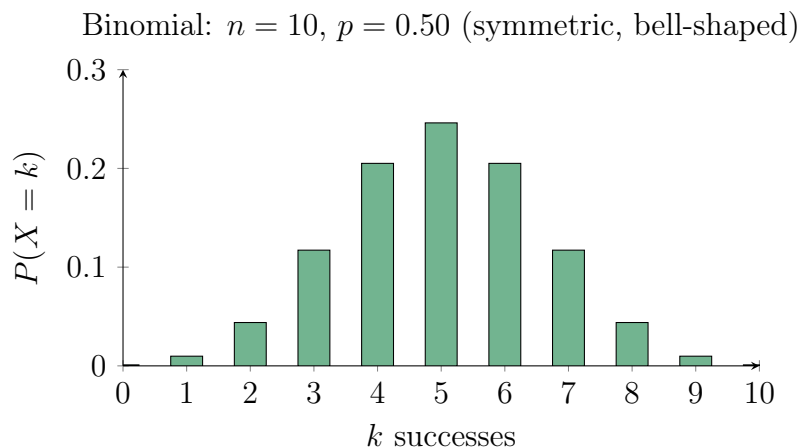
$$P(X = 10) = \binom{10}{10} (0.5)^{10} = 1/1024 \approx 0.0010$$

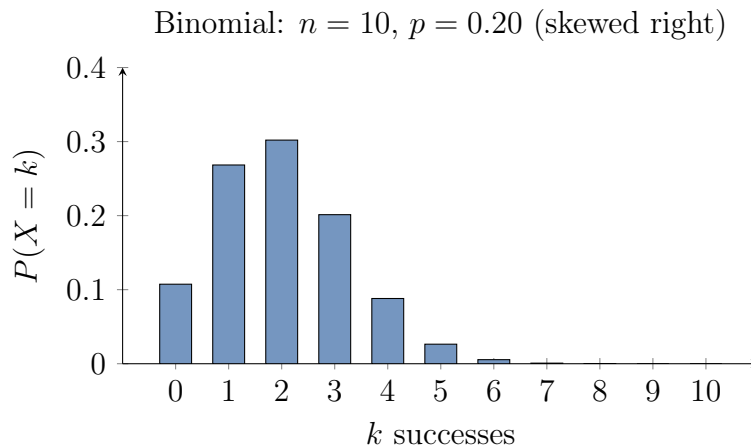
$$P(X \geq 7) \approx 0.1719.$$

About a 17.2% chance of passing by guessing alone.

2.4.4 The Shape of the Binomial Distribution

Visual Insight. The binomial histogram changes shape dramatically depending on p . When $p = 0.5$ and n is large, the histogram is strikingly bell-shaped and symmetric. When p is close to 0 or 1, it is skewed. This bell shape is no accident — it is the first glimpse of the normal distribution, which we will study in Chapter 3.





As p approaches 0.5 and n grows, the histogram fills in and approaches a smooth bell curve. The bars are becoming the curve. We will return to this idea in Chapter 4 when we study the Central Limit Theorem.

2.4.5 Mean and Standard Deviation of Binomial

Key Formula. For $X \sim \text{Bin}(n, p)$:

$$\mu = np, \quad \sigma = \sqrt{np(1-p)}.$$

These formulas have wonderful intuition. If you flip 100 fair coins, you expect $\mu = 100 \cdot 0.5 = 50$ heads. If you survey 200 people where 30% favor a policy, you expect $\mu = 200 \cdot 0.30 = 60$ favorable responses.

Example 2.4.4. (*Mean and Standard Deviation*) The true/false quiz: $n = 10, p = 0.5$.

$$\mu = 10(0.5) = 5, \quad \sigma = \sqrt{10(0.5)(0.5)} = \sqrt{2.5} \approx 1.58.$$

A student guessing randomly expects to get 5 correct, give or take about 1.6 questions. The center of mass of the histogram above is exactly at $k = 5$.

Common Pitfall. The binomial setting requires *independent* trials with the *same* p . If you are sampling without replacement from a small population, the trials are not independent (removing one item changes the probability for the next). In that case, the binomial formula gives an approximation. The approximation is good when the sample size is less than 10% of the population.

Exercises 2.4

2.4.1. A factory produces bolts where 5% are defective. A sample of 20 bolts is drawn. Let X = number of defective bolts. (a) Explain why this is (approximately) a binomial setting. (b) Compute $P(X = 0)$ and $P(X = 1)$. (c) Compute μ and σ . (d) Sketch the probability histogram for $k = 0, 1, 2, 3, 4, 5$. Describe its shape.

2.4.2. A multiple choice exam has 15 questions, each with 5 options. A student guesses randomly on every question. (a) What is the expected number of correct answers? (b) What is $P(X \geq 5)$? (Compute or explain how you would compute it.) (c) The pass mark is 9. Is passing by guessing realistic? Justify numerically.

2.4.3. For $X \sim \text{Bin}(n, 0.5)$: as n increases from 5 to 10 to 20, describe what happens to the shape of the histogram. What shape does it seem to approach? (You may sketch or describe.)

2.5 The Probability Histogram as a Unifying Picture

We close Chapter 2 by stepping back to see the big picture.

Every discrete distribution in this chapter can be drawn as a probability histogram:

- The **horizontal axis** shows the possible values.
- The **vertical axis** shows the probability of each value.
- The **bars all have width 1**, so *height equals area*.
- The **total area of all bars equals 1**.

Because bar height equals bar area (width is 1), probability is area. This is not just a notational convenience — it is the key that unlocks continuous distributions.

In Chapter 3, we will let the number of bars grow without bound while keeping total area equal to 1. The bars will become so narrow they are invisible, and their tops will trace a smooth curve. That curve is a **continuous probability distribution**. Probability will still be area — but the area under a curve instead of the area of bars.

Visual Insight. Think of it this way. Start with the binomial histogram for $n = 10$, $p = 0.5$. Now imagine $n = 50$, then $n = 200$. The bars get thinner and more numerous. The outline of the histogram smooths out. In the limit, you have a bell-shaped curve — the normal distribution.

Chapter 2 Review

1. A carnival game costs \$3 to play. You draw one card from a standard 52-card deck. If you draw an ace, you win \$20; if you draw a face card (J, Q, K), you win \$8; otherwise you win nothing. Let X be your net gain. (a) Write the probability distribution table. (b) Compute μ . Is the game in your favor? (c) Compute σ^2 and σ .
2. A quality control inspector examines circuit boards one at a time. Each board has a 4% defect rate. Boards are independent. (a) What is the probability the first defective board is the 5th one inspected? (b) How many boards does the inspector expect to check before finding a defect? (c) Now suppose the inspector checks exactly 25 boards and counts defectives. Is this geometric or binomial? Explain. (d) For part (c), compute $P(X = 0)$, $P(X = 1)$, and μ .
3. Sketch (or describe) the probability histograms for $\text{Bin}(10, 0.1)$ and $\text{Bin}(10, 0.9)$. How are they related to each other? Explain why.
4. *Big picture:* In your own words (2–3 sentences), explain why probability is described as “area” even in a discrete histogram where the outcomes are whole numbers. How does this connect to what you expect to see in Chapter 3?